"FAILURES GIVE YOU MUCH MORE UNDERSTANDING THAN SUCCESS"

The man of the moment, *ISRO chairman* DR S. SOMANATH, spoke at length to *Group Editorial Director* RAJ CHENGAPPA on the outcome of India's moon missions and beyond. Edited excerpts from the interview



Photograph by BANDEEP SINGH

Q. What do you see as the main achievements of Chandrayaan-3?

There are many. With a mission strategy to reach the moon at a comparatively slow pace and accomplish a soft landing while ensuring minimal cost is a very unique management plan that is intelligent, innovative and frugal. This is true of all the Chandrayaan launches. The second unique thing is that the systems that went into the making of the spacecraft were a great symbol of aatmanirbharta. It was all developed, designed and perfected in India, whether it was the electronics, the propulsion, the controls, the sensors or the other systems. The third thing was that we learnt from the limitations of the earlier two Chandrayaan missions because any failure gives you much more understanding than success does. We ensured that we learnt from them and implemented the solutions. For that, the decision-making and management process was critical and our approach included inclusiveness, openness to review, taking suggestions from everybody, including the larger scientific community beyond ISRO. Also insisting that there be no compromise while conducting all the tests, analysing performance, studying all suggestions thoroughly, addressing shortcomings and implementing corrections. I told the team we won't go for a launch until there was 100 per cent satisfaction at all levels.

> "Our mission was to achieve a soft landing while ensuring minimal cost. It was a unique, intelligent, innovative and frugal plan"

Q. Interestingly, you term this as a failure-based approach as opposed to the success-oriented one for the previous launch. What does that mean?

Let me clarify that a failure-oriented approach is not of pessimism. What it means is that we do not assume normal performance of all systems but instead plan for all things that can and will go wrong to make our boundaries for performance wider than we have ever thought about. Normally, we test everything to what is called 3-sigma level or three times the distribution and